

Project Objectives & Expected Outcomes

1. Core Objectives

1. Build a robust Machine Learning pipeline that predicts human development classifications with high accuracy ($\geq 98\%$).
2. Develop an interactive Scenario Simulator that allows users to adjust indicator sliders (Life Expectancy, Schooling, GNI per Capita) and receive instant, real-time predictions.
3. Incorporate Explainable AI (XAI) algorithms to describe the drivers behind each classification, alongside a recommendation engine.
4. Implement client-side PDF Report compilation and CSV dataset exports to simplify data distribution.

2. Expected Outcomes

1. Sandbox Tool: Provides a functional sandbox for testing simulated national investments.
2. Actionable Policy Guidelines: Automates recommendations based on calculated gaps between baseline indicators and top-performing country benchmarks.
3. Portfolio Showcase: Displays clean coding principles, SPA design, unit testing, and robust Flask backend patterns.